Dataset & Baseline – Week 2

Group	Group 2 - AI Exam Answer Grader
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Class & Course	B.Tech DS 4th year – Speech & Natural Language Processing (SNLP)

1. Dataset

For the project, we have used the EngSAF dataset (https://huggingface.co/datasets/IsmaelMousa/engsaf). The file '*load_data.ipynb*' executes the loading of the training, validation and testing datasets from HuggingFace. Then, it carries out basic cleaning on the student answer and reference answers to save them in the /data folder.

The dataset has 4735, 338 and 506 samples in training, validation and test splits. The dataset originally has questions, student_answer, reference_answer, mark_scheme, score, rationale as the columns.

```
dataset = load_dataset("IsmaelMousa/engsaf")
print(dataset)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
DatasetDict({
    train: Dataset({
        features: ['question', 'student_answer', 'reference_answer', 'mark_scheme', 'score', 'rationale'],
        num_rows: 4735
    })
    validation: Dataset({
        features: ['question', 'student_answer', 'reference_answer', 'mark_scheme', 'score', 'rationale'],
        num_rows: 338
    })
    test: Dataset({
        features: ['question', 'student_answer', 'reference_answer', 'mark_scheme', 'score', 'rationale'],
        num_rows: 506
    })
})
```

2. Baseline Model

For the Baseline Model, TF-IDF is used along with logistic regression to classify the scores as shown in the file '*baseline_tf_idf.ipynb*'.

It loads the preprocessed data splits and applies a TF-IDF vectorizer to transform the student and reference answers into numerical matrices. The pipeline then calculates the cosine similarity between these vectors and uses a Logistic Regression classifier to calibrate this continuous similarity score against the dataset's discrete ordinal ground truth labels.

The baseline model achieves: 50.29% training accuracy, 50.59% validation accuracy and 50.20% testing accuracy.

For the baseline model, we get the following confusion matrix and classification report upon test set as:

```
Confusion Matrix (Test):
[[ 48  18  47]
 [ 26  46 119]
 [ 16  26 160]]

Classification Report (Test):
              precision    recall  f1-score   support

           0       0.53      0.42      0.47       113
           1       0.51      0.24      0.33       191
           2       0.49      0.79      0.61       202

    accuracy                           0.50       506
   macro avg       0.51      0.49      0.47       506
weighted avg       0.51      0.50      0.47       506
```